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A comparison of the fine saccharide-binding specificity of *Dioclea grandiflora* lectin and concanavalin A

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The lectin from the seeds of *Dioclea grandiflora* (DGL) is a Man/Glc-specific tetrameric protein with physical and saccharide-binding properties reported to be similar to that of the jack bean lectin concanavalin A (ConA). Unlike other plant lectins, both DGL and ConA bind with high affinity to the core trimannoside moiety, 3,6-di-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside, which is present in all asparagine-linked carbohydrates. In the present study, hemagglutination inhibition techniques have been used to investigate binding of DGL and ConA to a series of mono- and dideoxy analogs of methyl 3,6-di-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside and to a series of asparagine-linked oligomannose and complex oligosaccharides and glycopeptides. The results indicate that both DGL and ConA recognize epitopes on all three residues of the trimannoside: the 3-, 4-, and 6-hydroxyl groups of the α(1–6)Man residue, the 3-hydroxyl group of the α(1–3)Man residue, and the 2- and 4-hydroxyl groups of the central Man residue of the core trimannoside. However, unlike ConA, DGL does not bind to biantennary complex carbohydrates. This was confirmed by showing that biantennary complex glycopeptides do not bind to a DGL-Sepharose affinity column. Unlike ConA, DGL does not show enhanced affinity for a large N-linked oligomannose carbohydrate (Man9 glycopeptide) relative to the trimannoside. Thus, DGL and ConA share similar epitope recognition of the core trimannoside moiety. However, they exhibit differences in their fine specificities for larger N-linked oligomannose and complex carbohydrates.

Keywords: lectin; specificity; cross-linking; carbohydrate; glycopeptide.

The seed lectin from *Dioclea grandiflora* (DGL) is a Man/Glc-binding protein isolated from the plant found in North East Brazil. The lectin is devoid of covalently linked carbohydrate, and is reported to be a tetramer with a molecular mass of 100 kDa (Moreira et al., 1983). Like other legume lectins, DGL requires Ca^{2+} and Mn^{2+} for its binding activity. The lectin possesses a high degree of sequence similarity with the jack bean lectin concanavalin A (ConA), another member of the *Dioclea* tribe, and differs in only 52 out of 237 residues (Richardson et al., 1984). Six of the seven residues that have been implicated as ligands for Ca^{2+} and Mn^{2+} are conserved, and the seven amino acid residues surrounding the shallow saccharide-binding site of ConA are also conserved in DGL (Richardson et al., 1984).

Unlike other Man-specific plant lectins, DGL and ConA bind with high affinity to the core trimannoside moiety, 3,6-di-*O*-(α-D-mannopyranosyl)-α-D-mannopyranose, which is present in all N-linked carbohydrates (Brewer and Bhattacharyya, 1986; Chervenak and Toone, 1995). To investigate the binding epitopes recognized by DGL and ConA for the trisaccharide, a series of mono- and dideoxy derivatives of methyl 3,6-di-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside (**16**, Fig. 1) were synthesized and tested for their ability to bind to both lectins as determined by hemagglutination inhibition methods. In addition,

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Abbreviations. DGL, seed lectin from *Dioclea grandiflora*; Con A, *concanavalin A* lectin from jack bean; α-ManMe, methyl α-D-mannopyranoside; α-2dManMe, methyl α-D-2-deoxymannopyranoside; α-GlcMe, methyl α-D-glucopyranoside; β-GlcMe, methyl β-D-glucopyranoside; dMan, deoxy-D-mannose; **16**, methyl 3,6-di-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside (see Fig. 1 for structures of **16**–**32**); **17**, methyl 6-*O*-(α-D-glucopyranosyl)-3-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside; **18**, methyl 6-*O*-(α-D-galactopyranosyl)-3-*O*-(α-mannopyranosyl)-α-D-mannopyranoside; **19**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(α-D-glucopyranosyl)-α-D-mannopyranoside; **20**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(α-D-galactopyranosyl)-α-D-mannopyranoside; **21**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(2-deoxy-α-D-mannopyranosyl)-α-D-mannopyranoside; **22**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(3-deoxy-α-D-mannopyranosyl)-α-D-mannopyranoside; **23**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(4-deoxy-α-D-mannopyranosyl)-α-D-mannopyranoside; **24**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(6-deoxy-α-D-mannopyranosyl)-α-D-mannopyranoside; **25**, methyl 6-*O*-(2-deoxy-α-D-mannopyranosyl)-3-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside; **26**, methyl 6-*O*-(3-deoxy-α-D-mannopyranosyl)-3-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside; **27**, methyl 6-*O*-(4-deoxy-α-D-mannopyranosyl)-3-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside; **28**, methyl 6-*O*-(6-deoxy-α-D-mannopyranosyl)-3-*O*-(α-D-mannopyranosyl)-α-D-mannopyranoside; **29**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(α-D-mannopyranosyl)-2-deoxy-α-D-mannopyranoside; **30**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(α-D-mannopyranosyl)-4-deoxy-α-D-mannopyranoside; **31**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(3-deoxy-α-D-mannopyranosyl)-2-deoxy-α-D-mannopyranoside; **32**, methyl 6-*O*-(α-D-mannopyranosyl)-3-*O*-(3-deoxy-α-D-mannopyranosyl)-4-deoxy-α-D-mannopyranoside; NMRD, nuclear magnetic relaxation dispersion, the magnetic field dependence of nuclear magnetic relaxation rate (longitudinal) of solvent protons; CHO, Chinese hamster ovary.

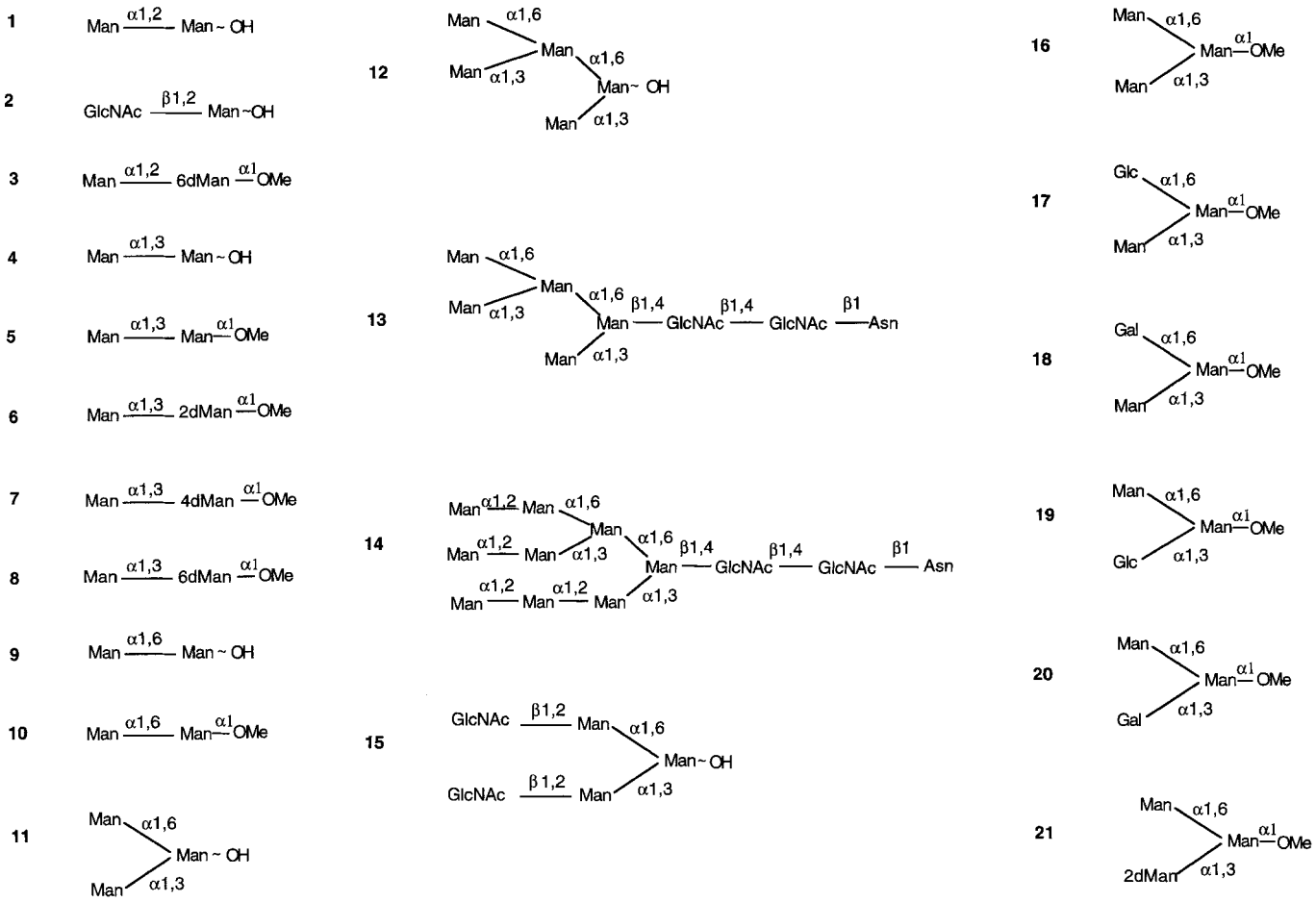

Fig. 1. Structures of oligosaccharides 1–33. Man, GlcNAc, Gal, Glc, 2dMan, 3dMan, 4dMan, 6dMan and Asn represents mannose, *N*-acetylglucosamine, galactose, glucose, 2-deoxymannose, 3-deoxymannose, 4-deoxymannose, 6-deoxymannose and asparagine, respectively.

both lectins were tested for their ability to bind to certain N-linked oligomannose and complex oligosaccharides and glycopeptides. The results of these studies reveal that both ConA and DGL have similar extended binding sites that recognize the same epitopes in the trimannoside. However, important differences exist in their binding specificities for larger N-linked oligomannose and complex type carbohydrates and glycopeptides.

MATERIALS AND METHODS

ConA was prepared from jack bean seeds (Sigma) according to the method of Agarwal and Goldstein (Agrawal and Goldstein, 1967). The concentration of ConA was determined spectrophotometrically at 280 nm using $A^{1\%}_{1cm}$ = 13.7 at pH 7.2 (Goldstein and Poretz, 1986) and expressed in terms of the monomer molecular mass 25600 Da. DGL was isolated from *Dioclea grandiflora* seeds obtained from North Eastern Brazil (Albano Ferreira Martin Ltd) as previously described (Moreira et al., 1983). The concentration of DGL was determined spectrophotometrically at 280 nm using $A^{1\%}_{1cm}$ = 12.0 at pH 7.2 and expressed in terms of the monomer (25600) (Moreira et al., 1983). α-GlcMe, α-ManMe, β-GlcMe, 2dMan, GlcNAc, Gal, **5**, **10**, **16** (Fig. 1 and Abbreviations section), maltose, and maltotriose were purchased from Sigma. α-2dManMe was a gift from Dr S. Sabesan, Du Pont Company, Wilmington, DE. [^{3}H]Glc is D-[6-^{3}H]Glc (36.25 Ci/mmol) from Amersham. **1**, **2**, **4**, **9**, **11**, **12** and **15** (Fig. 1) were obtained from Dextra Laboratories Ltd. Oligosaccharides **17–20** (Fig. 1) were synthesized as described earlier (Garegg et al., 1990). The synthesis of oligosaccharides **3**, **6–8**, **21–32** (Fig. 1) has been described (Oscarson and Tedebark, 1995). Man5 glycopeptide (**13**) was obtained from hen egg white ovalbumin, and Man5 oligosaccharide **12** (Fig. 1) was purchased from Dextra Laboratories. Man9 glycopeptide (**14**) was prepared from a pronase digest of the soybean agglutinin as previously described (Lis and Sharon, 1978). Standard oligomannosyl glycopeptides were prepared from chicken egg ovalbumin according to Huang and coworkers (Huang et al., 1970). Biantennary complex glycopeptides were isolated from human fibrinogen and human IgG as described earlier (Stanley, 1987). The concentrations of carbohydrates were measured by the phenol-sulfuric acid method using appropriate mixture of Man, Glc, and Gal as the standard (Dubois et al., 1956; Saha and Brewer, 1994). Purity of the oligosaccharides was checked by 500-MHz ^{1}H-NMR spectroscopy. Pronase (*Streptomyces grisens*) was from Boehringer Mannheim. ConA-Sepharose was from Pharmacia. DGL-Sepharose was prepared by the cyanogen bromide activation method (March et al., 1974). Ecolume was from ICN Biochemicals.

Hemagglutination inhibition assay. The assay was performed at 22 °C using a twofold serial dilution technique (Osawa and Matsumoto, 1972) and 3% (by vol.) rabbit erythrocytes in Hepes buffer (0.1 M Hepes, 0.15 M NaCl, 1 mM $CaCl_2$ and 1 mM $MnCl_2$, pH 7.2). The minimum concentration of saccharide required for complete inhibition of four hemagglutination doses was determined.

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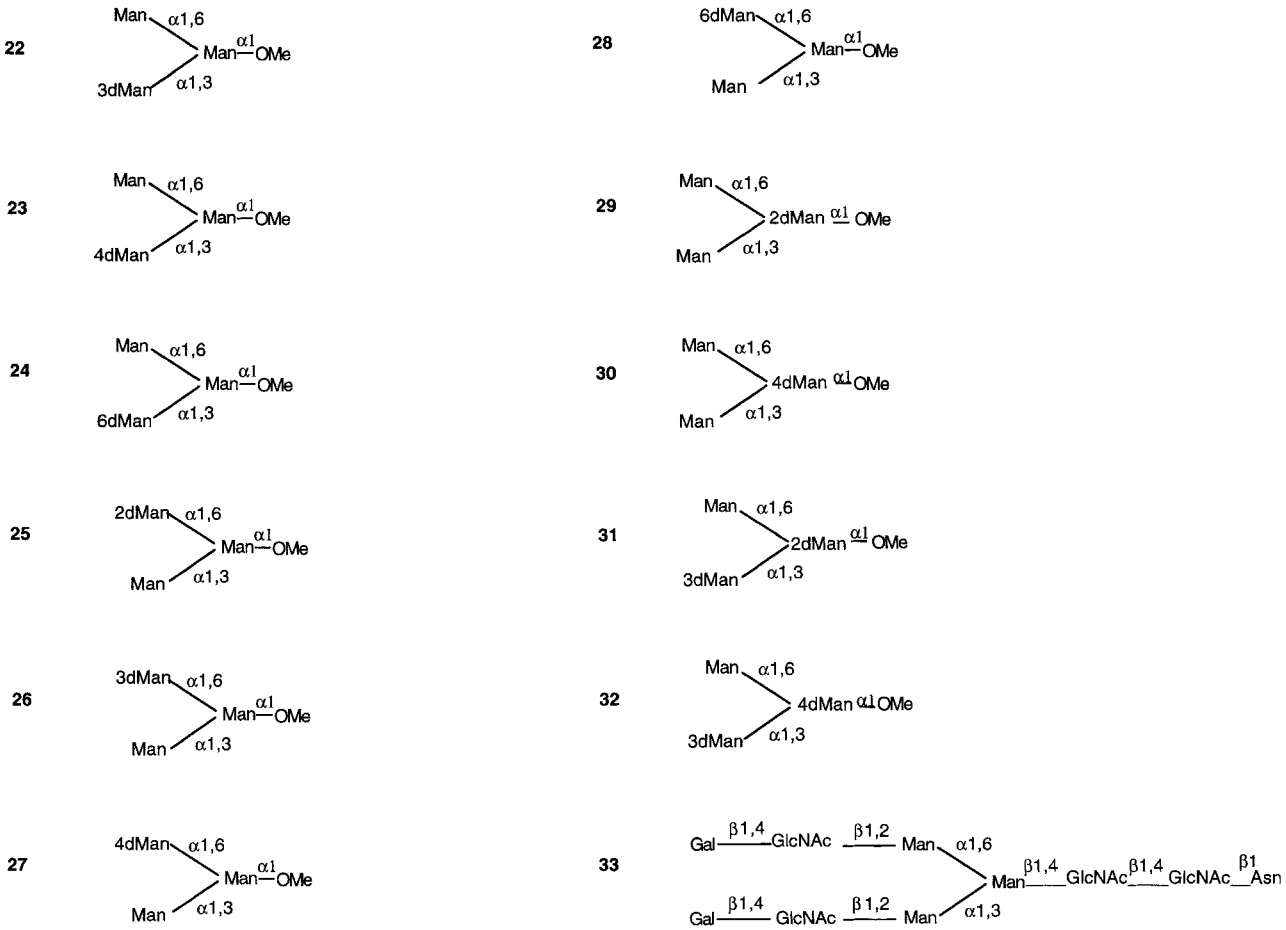

Fig. 1. Continued.

Quantitative precipitation studies. Quantitative precipitation profiles were performed in 100 µl 0.1 M Hepes buffer, pH 7.2, containing 0.9 M NaCl, 1 mM $CaCl_2$, and 1 mM of $MnCl_2$ as described previously (Mandal and Brewer, 1992). The formation of precipitates was inhibited or precipitates were dissolved by addition of 0.1 M Man. Nonspecific binding sugars such as Glc or Fuc had no effect.

Isolation and purification of radiolabelled cell surface glycopeptides. Chinese hamster ovary (CHO) cells were grown in suspension at 37 °C in complete alpha medium (Gibco) containing 10% fetal calf serum as described previously (Stanley, 1983). A suspension culture of Pro-5 parental CHO cells was established at a density of 1.5×10^5 cells/ml in complete medium and 100 µCi [^{3}H]Glc was added to each 10-ml culture. Following a 48-h incubation at 37 °C, the cells were washed three times with 10 ml NaCl/P_i (20 mM sodium phosphate, 150 mM sodium chloride, pH 7.4) containing 10 mM $CaCl_2$, 10 mM $MgCl_2$, and 0.02% azide. The cells were treated with 5 mM Tris/HCl, pH 7.8, containing 1% NP-40 for 5 min at 4 °C and 5 min at room temperature. Cells were then centrifuged and the supernatant was incubated with 5 mM Tris/HCl, pH 7.8, containing 1 mg/ml pronase and 3 mM $CaCl_2$ at 55 °C under toluene. Additional pronase was added every 48 h for 3 days. The reaction was stopped by boiling the reaction mixture for 10 min and the mixture was centrifuged for 10 min at 12000 *g*, desalted on Bio-Gel P-2, and stored at −20 °C until further use.

Lectin affinity chromatography. The radiolabelled glycopeptides were applied at room temperature to ConA-Sepharose or DGL-Sepharose columns in 0.1 M sodium acetate, pH 7.3, containing 10 mM $MgCl_2$, 10 mM $CaCl_2$, 10 mM $MnCl_2$, and 0.02% azide. The column was washed with the same buffer and glycopeptides were sequentially eluted with 10 mM α-GlcMe, followed by 10 mM α-ManMe and finally with 200 mM α-ManMe in the same buffer. Fractions (1 ml) were collected and mixed with Ecolume and counted in a β-scintillation counter.

RESULTS AND DISCUSSION

Sequence analysis of the α-subunit of DGL demonstrates that it is a single polypeptide chain of 237 amino acids that differs from the sequence of ConA in only 52 positions (Richardson et al., 1984). Six of the seven residues implicated as protein ligands for Mn^{2+} and Ca^{2+} in ConA (Glu8, Tyr12, Asn14, Asp19, and His24) are conserved in DGL. The monosaccharide-binding site of ConA is a shallow pocket located about 0.7 nm from the Ca^{2+} and 1.2 nm from the Mn^{2+} site surrounded by Asn14, Asp16, Asp208, Tyr12, Tyr100, Leu99, and Arg228 (Hardman and Ainsworth, 1976; Derewenda et al., 1989). All of these amino acid residues are conserved in DGL (Richardson et al., 1984). These results indicate that the binding specificity of both lectins for monosaccharides should be very similar. The results in Table 1 confirm these expectations.

Monosaccharide-binding specificities. Previous studies have shown that DGL and ConA bind to Man and Glc but not to Gal (Moreira et al., 1983; Goldstein and Poretz, 1986). DGL, like ConA, is also reported to selectively bind to α-anomeric glycosides (Chervenak and Toone, 1995). These findings are confirmed by the data in Table 1. Interestingly, 2dMan and α-

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Table 1. Relative inhibitory potencies of monosaccharides and disaccharides for DGL- and ConA-mediated hemagglutination of rabbit erythrocytes. Data in parentheses are relative inhibitory potencies normalized to α-ManMe for each lectin. The assay was performed at 22°C using a twofold serial dilution technique and 3% (by vol.) rabbit erythrocytes in Hepes buffer (0.1 M Hepes, 0.15 M NaCl, 1 mM $CaCl_2$ and 1 mM $MnCl_2$, pH 7.2). The minimum concentration of saccharide required for complete inhibition of four hemagglutination doses was determined. n.i., no inhibition.

Saccharide	Minimum concentration required for complete inhibition	
	DGL	ConA
	mM	
Man	15 (0.2)	15 (0.2)
α-ManMe	3.1 (1)	3.1 (1)
β-GlcMe	n.i. at 150 mM	n.i. at 150 mM
α-GlcMe	25 (0.12)	13 (0.25)
2dMan	n.i. at 100 mM	25 (0.12)
α-2dManMe	n.i. at 100 mM	9.3 (0.34)
GlcNAc	n.i. at 100 mM	n.i. at 50 mM
Gal	n.i. at 100 mM	n.i. at 100 mM
Maltose	n.i. at 50 mM	25 (0.12)
Maltotriose	50 (0.06)	25 (0.12)
3dMan	n.i. at 200 mM	n.i. at 200 mM
4dMan	n.i. at 200 mM	n.i. at 200 mM
6dMan	n.i. at 200 mM	n.i. at 200 mM
1	4.6 (0.68)	0.60 (5.4)
2	n.i. at 104 mM	6.5 (0.48)
3	1.07 (2.9)	0.54 (5.8)
4	3.1 (1.01)	3.1 (1.01)
5	0.97 (3.2)	0.97 (3.2)
6	3.3 (0.94)	3.3 (0.94)
7	1.1 (2.8)	0.24 (12.8)
8	0.42 (7.5)	0.42 (7.5)
9	2.0 (1.58)	2.0 (1.58)
10	6.1 (0.51)	6.0 (0.52)
11	0.012 (260)	0.025 (125)

Table 2. Relative inhibitory potencies of oligosaccharides and glycopeptides for DGL- and ConA-mediated hemagglutination of rabbit erythrocytes. Data in parentheses are relative inhibitory potency normalized to **11** for each lectin. The assay was performed at 22°C using a twofold serial dilution technique and 3% (by vol.) rabbit erythrocytes in Hepes buffer (0.1 M Hepes, 0.15 M NaCl, 1 mM $CaCl_2$ and 1 mM $MnCl_2$, pH 7.2). The minimum concentration of saccharide required for complete inhibition of four hemagglutination doses was determined. n.i., no inhibition.

Saccharide	Minimum concentration required for complete inhibition	
	DGL	ConA
	μM	
11	12 (1)	25 (1)
12	6.0 (2)	27 (0.93)
13	7.0 (1.70)	28 (0.89)
14	3.0 (4)	1.0 (25)
15	400 (0.03)	6.0 (4.2)
16	7 (1.70)	28 (0.89)
17	59 (0.20)	120 (0.21)
18	760 (0.02)	760 (0.03)
19	350 (0.03)	880 (0.03)
20	180 (0.07)	1000 (0.03)
21	6.0 (2)	25 (1)
22	140 (0.08)	290 (0.09)
23	9 (1.33)	26 (0.96)
24	6 (2)	32 (0.78)
25	36 (0.33)	36 (0.69)
26	400 (0.03)	200 (0.13)
27	830 (0.01)	230 (0.11)
28	580 (0.02)	240 (0.10)
29	59 (0.20)	59 (0.42)
30	61 (0.20)	120 (0.21)
31	750 (0.02)	1500 (0.02)
32	180 (0.07)	350 (0.07)
33	n.i. at 1.6 mM	200 (0.13)

2dManMe both inhibit ConA but show no activity against DGL. GlcNAc shows no binding to ConA and DGL, under these conditions. ConA has previously been shown by Goldstein and coworkers (Goldstein and Poretz, 1986) to recognize the 3-, 4-, and 6-hydroxyl groups of Man. The lack of inhibition by 3dMan, 4dMan, and 6dMan for DGL indicates a similar requirement for binding of these hydroxyl groups.

Disaccharide-binding specificities. Maltose and maltotriose which bind to ConA show only weak interactions, if any, with DGL (Table 1). Another difference between the two lectins is the higher affinity of ConA for the α(1−2) disaccharide of Man (**1**) (nearly eightfold higher than DGL), which reflects the well known enhanced affinity of ConA for α(1−2) di- and trisaccharides of Man (So and Goldstein, 1968). Titration microcalorimetry studies have shown that the enhanced binding of α(1−2) di- and trisaccharides of Man for ConA is due, primarily, to more favorable entropy interactions, relative to the binding of oligosaccharides with different glycosidic linkages (Mandal et al., 1994a; Chervenak and Toone, 1995). DGL, however, does not exhibit enhanced binding for **1**. The difference in affinity of DGL and ConA for **1** is important for understanding differences in their affinities for the Man9 glycopeptide (**14**) (Table 1), as discussed below. The data for **3**, which is an analog of **1** that possesses a 6dMan residue at the reducing terminal, indicate that the absence of a 6-hydroxyl group in this position enhances binding to DGL by a factor of four, whereas binding to ConA is unaffected.

The relative binding affinities of DGL and ConA to α(1−3) dimannosides **4** and **5** as well as to monodeoxy analogs **6** and **8** are similar. The same is true for α(1−6) dimannosides **9** and **10**, which are similar. These observations confirm recent titration microcalorimetry results for the binding of DGL and ConA to **5** and **10**, respectively (Chervenak and Toone, 1995). Oligosaccharide **7** which possesses a 4dMan residue at the reducing terminal of α(1−3) dimannoside **5** shows nearly 13-fold enhanced affinity for ConA, relative to the binding of α-ManMe, but only a threefold enhanced affinity for DGL. These results indicate subtle differences in the binding sites of the two lectins for α(1−3) dimannosides.

A substantial difference in the binding specificities of the two lectins is observed for the disaccharide GlcNAc(β1−2)Man (**2**). Table 1 shows that ConA binds to **2** nearly as well as to α-ManMe, but that DGL does not bind to **2**. This result is important since **2** is part of the structure of complex-type oligosaccharides which are also distinguished in their interactions with the two lectins, as discussed below.

Binding of core trimannoside (11) and its analogs. Table 2 shows the relative binding affinities of DGL and ConA for the core trimannoside (**11**), and for Glc, Gal, and mono- and dideoxy analogs of the trimannoside. Table 1 shows that both DGL and ConA bind with high affinity to **11**, relative to the binding of

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α-ManMe, as previously reported (Brewer and Bhattacharyya, 1986; Chervenak and Toone, 1995). Since DGL and ConA bind well to the methyl α-anomer of the trimannoside (**16**), analogs of the methyl α-anomer of the trimannoside were synthesized possessing either Glc or Gal substituted at either nonreducing terminal residue, and mono- and dideoxy analogs of Man at all three residues.

Analogs **17** and **19** contained Glc substituted at the α(1−6) and α(1−3) positions, respectively, in the trimannoside. Analogs **18** and **20** contained Gal substituted at the α(1−6) and α(1−3) positions, respectively. Previous studies have shown that these analogs all show reduced affinity for ConA (Mandal et al., 1994b). Table 2 shows that the relative binding affinities of DGL and ConA for analogs **17–20** are very similar. DGL and ConA both exhibit the highest affinity for **17** out of the four analogs. This finding is in agreement with the earlier study of Mandal and coworkers (Mandal et al., 1994b), which concluded that the Glc residue of **17** binds to the so-called monosaccharide-binding site of ConA based on the loss in affinity of **17** for the lectin relative to **16**. These results indicate that DGL and ConA recognize all four analogs in the same manner, which suggests a highly conserved binding site for the trimannoside for both proteins. The results also suggest, by analogy with ConA, that the Glc moiety of **17** occupies the monosaccharide-binding site of DGL.

Table 2 shows the relative affinities of DGL and ConA for mono- and dideoxy analogs of **16**. A previous study demonstrated that monodeoxy analogs of the α(1−3) arm of the trimannoside (**21−24**) showed no loss of binding to ConA except the 3dMan analog (**22**) (Mandal et al., 1994b). Titration microcalorimetry measurements indicated an order of magnitude loss in affinity of ConA for **22**, and a loss of 14.3 kJ mol^{-1} in its enthalpy of binding relative to the binding of **16**. Table 2 shows that DGL like ConA also exhibits a similar pattern of binding to **21–24** in that DGL demonstrates a loss of binding only to **22** of about the same degree exhibited by ConA. This indicates conservation of this epitope in **16** for both lectins. The recent X-ray crystal structure of ConA complexed with **16** shows hydrogen bonding of the 3-OH on the α(1−3) arm of the trimannoside to the lectin (Naismith and Field, 1996).

The synthesis of monodeoxy analogs of the α(1−6) arm of the trimannoside (**25−28**) affords the opportunity of determining the effects of such modifications in the oligosaccharide on its binding to DGL and ConA. Table 2 shows that ConA exhibits essentially the same affinity for the 2dMan analog (**25**) as for the native trimannoside (**16**). However, ConA shows an order of magnitude loss in affinity for the 3- (**26**), 4- (**27**), and 6-deoxy (**28**) analogs. These findings are similar to those for the binding of deoxy analogs of Man, a monosaccharide (Table 1), as previously reported by Goldstein and coworkers (Goldstein and Poretz, 1986). Thus, the results for **25-28** indicate that the α(1−6) arm of the trimannoside occupies the so-called monosaccharide-binding site of ConA, as previously suggested (Mandal et al., 1994b). This is confirmed by the recent X-ray crystal structure of ConA bound to **16** (Naismith and Field, 1996). The same pattern of binding of DGL to **25−28** is apparent for the data in Table 2. There is a greater loss in binding of DGL to the 2dMan analog (**25**) relative to ConA, consistent with the greater loss in binding of DGL to 2dMan relative to ConA (Table 1). DGL shows even greater loss in binding, relative to ConA, to the 3- (**26**), 4- (**27**), and 6-deoxy (**28**) analogs. These results, like the results for ConA, indicate that the α(1−6) arm of the trimannoside occupies the so-called monosaccharide-binding site of DGL. Thus, both lectins bind to **16** with its α(1−6) Man residue occupying the monosaccharide-binding sites of both proteins.

Oligosaccharide analogs **29** and **30** possess 2dMan and 4dMan residues substituted at the central Man residue of the trimannoside, respectively. DGL shows a fivefold loss in binding to both **29** and **30**, with respect to **16**, which indicates a requirement for binding to the 2- and 4-hydroxyl groups of the central Man residue of the trimannoside. ConA shows over a twofold loss in binding to **29** and a fivefold loss in binding **30**. Titration microcalorimetry measurements also confirm losses in binding of ConA to **29** and **30** (Brewer, C. F., unpublished results). A previous titration microcalorimetry study suggests the involvement of the central Man residue of **16** in binding to ConA based on the affinity of the α(1−3) 3dMan analog (**22**) which was greater than that of α-ManMe (Mandal et al., 1994b). These results indicate that ConA possesses extended site interactions with not only the 3-hydroxyl group of the α(1−3) Man residue of the trimannoside, but also with the 2- and 4-hydroxyl groups of the central Man residue in addition to the 3-, 4-, and 6-hydroxyl groups of the α(1−6) Man residue. These findings are consistent with the recent X-ray crystal structure of ConA bound to **16** which shows hydrogen-bonding interactions of these two hydroxyl groups of the central Man residue with the protein (Naismith and Field, 1996). The present data indicate a remarkable conservation of these epitopes of the trimannoside that are required for binding to DGL.

Dideoxy analogs **31** and **32** which possess the α(1−3) 3dMan residue as well as the 2dMan and 4dMan central residue of the trimannoside, respectively, show even greater losses in binding to DGL and ConA with respect to the corresponding monodeoxy analogs (Table 2). These results further support the involvement of these hydroxyl groups of the trimannoside in binding to the two lectins.

Binding of N-linked oligomannose and complex carbohydrates. ConA is known to bind with high affinity to N-linked oligomannose and complex carbohydrates (Brewer and Bhattacharyya, 1986; Bhattacharyya et al., 1987a, b; Bhattacharyya and Brewer, 1989). The results in Table 2 show that ConA binds to the Man5 oligosaccharide (**12**) and Man5 glycopeptide (**13**) with essentially the same affinity as trimannoside **11** (and **16**), as previously reported (Brewer and Bhattacharyya, 1988). This demontrates that the trimannoside moiety on the α(1−6) arm(s) of **12** and **13** is the primary epitope for ConA binding. ConA is also known to bind with considerable higher affinity to the Man9 glycopeptide (**14**) (25-fold, Table 2) (Bhattacharyya and Brewer, 1989). The higher affinity of the glycopeptide has been explained in terms of the presence of the α(1−2) linked Man residues on both the α(1−6) and α(1−3) arms, which together with the trimannoside moiety on the α(1−6) arm, all contribute to binding. The α(1−2)−linked Man residues are able to bind and slide back and forth between adjacent Man residues in the chain since the 3-, 4-, and 6-hydroxyl groups of both terminal and internal residues are available for binding to the lectin (Bhattacharyya and Brewer, 1989). This results in a decrease in the entropy of binding of ConA to α(1−2) Man oligosaccharides and increase in their affinity relative to α(1−3)-, α(1−4)-, and α(1−6)-linked oligosaccharides (Mandal et al., 1994a). ConA, for example, binds to the α(1−2) Man disaccharide (**1**) with fivefold increased affinity relative to α-ManMe (Table 1) (So and Goldstein, 1968; Mandal et al., 1994a).

DGL also binds to the Man5 oligosaccharide (**12**) and Man5 glycopeptide (**13**) with the same affinity as **16** (Table 2). However, DGL shows only a twofold increase in affinity for the Man9 glycopeptide (**14**) relative to the trimannoside. This lack of enhanced binding of DGL to **14** appears related to the lack of enhanced binding of DGL to α(1−2) Man disaccharide (**1**) (Table 1). Unlike ConA, DGL appears unable to bind to both

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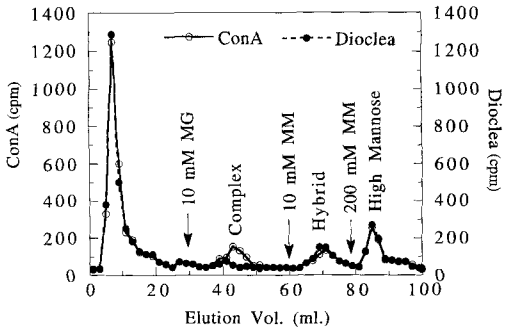

Fig. 2. Lectin affinity chromatography of glycopeptides from CHO. Radiolabelled glycopeptides from CHO cells were similarly fractionated on ConA-Sepharose and DGL-Sepharose columns. The bound glycopeptides were eluted as described in the Materials and Methods section. Standard glycopeptides of the biantennary complex, hybrid and oligomannosyl (high mannose) classes elute from ConA-Sepharose and DGL-Sepharose at the positions shown. MG, α-GlcMe; MM, α-ManMe.

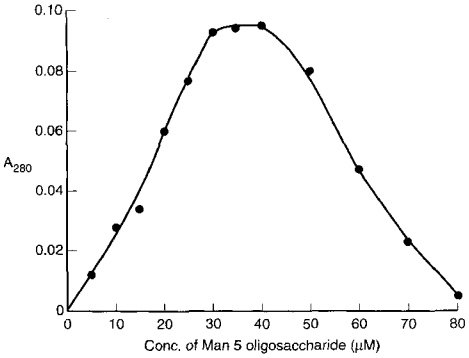

Fig. 3. Quantitative precipitation profile of DGL with oligosaccharide 12 at 22 °C. The pH 7.2 buffer contained 0.1 M Hepes, 0.9 M NaCl, 1 mM $CaCl_2$ and 1 mM $MnCl_2$. The concentration of DGL was 60 μM.

Man residues in α(1−2) Man and to slide from one to another. Thus, although DGL recognizes the trimannoside moiety on the α(1−6) arm of **14**, it does not possess enhanced affinity due to the presence of the α(1−2) Man residues in the glycopeptide.

Another more striking difference in the binding properties of DGL and ConA is their relative affinities for biantennary complex-type carbohydrates **15** and **33**. Oligosaccharide **15** possesses an added β(1−2) GlcNAc residue on each arm of **11**, while **33** possesses added β(1−2) LacNAc residues. Although ConA binds to **15** and **33** (Table 2) (Bhattacharyya et al., 1987b), DGL shows only very weak interactions with **15** and essentially no interactions with **33** at high concentrations of the latter (1.6 mM). These results are directly related to lack of binding of DGL to disaccharide **2** which is present in **15** and **33**. In the case of **2**, the absence of DGL binding appears to be due to its lack of binding to a nonreducing β-GlcNAc residue and inability to bind to the reducing Man residue. In the case of **15**, the presence of the outer β(1−2) GlcNAc residue(s) appears to interfere with binding to the internal trimannoside moiety. In the case of **33**, the presence of the nonreducing β-Gal residue(s) appears to interfere even more with DGL binding. Thus, DGL, unlike ConA, shows no ability to bind to biantennary complex carbohydrates. This difference in the binding specificities of ConA and DGL was further demonstrated by subjecting a mixture of high mannose, hybrid and complex glycopeptides isolated from CHO cells to affinity chromatography on ConA-Sepharose and DGL-Sepharose affinity columns (Fig. 2). High mannose and hybrid glycopeptides bound to both ConA and DGL affinity columns and eluted at the same positions, whereas biantennary complex glycopeptides bound to the ConA column but not to the DGL column. These results confirmed the unique ability of DGL to bind to high mannose and hybrid glycopeptides but not to biantennary complex carbohydrates.

The combined lack of enhanced affinity of DGL for Man(α1−2)Man (**1**), relative to α-ManMe, and lack of binding to Glc(β1−2)Man (**2**) strongly suggest that, unlike ConA, DGL does not bind to internal or reducing terminal Man or Glc residues with free 3-, 4-, and 6-hydroxyl groups in oligosaccharides. The exception appears to be binding of DGL to the interior trimannoside moiety on the α(1−6) arm in the Man9 glycopeptide (**14**). In contrast, DGL does not bind to the internal trimannoside moiety in **15** and **33**. Thus, DGL, unlike ConA, is sensitive to the nature of the linkage and/or saccharides residues adjacent to the trimannoside moiety in different N-linked carbohydrates, which accounts for its binding to oligomannose glycopeptides such as **14** but not to complex-type carbohydrates like **15** and **33**. Since the X-ray crystal structure of ConA complexed with **16** has been determined (Naismith and Field, 1996), it will be interesting to compare the X-ray crystal structure of both lectins to determine the structural basis of the fine differences in their binding specificities.

Quantitative precipitation studies of DGL with 12 and 14. Quantitative precipitation experiments show that DGL precipitates with **12** and **14** under appropriate conditions, similar to other lectin-carbohydrate precipitation profiles (Bhattacharyya et al., 1987a, 1988, 1989; Bhattacharyya and Brewer, 1988). Fig. 3 shows the quantitative precipitation curve for DGL (60 μM) in the presence of increasing concentrations of **12** at pH 7.2. The accumulation of precipitate could be prevented in the presence of 0.1 M MeαMan, the precipitate was dissolved upon addition of the monosaccharide. The concentration of oligosaccharide at the equivalence point is 35 μM. Thus, the ratio of the concentration of oligosaccharide to the protein monomer at the equivalence point (maximum precipitation) of the precipitin curve is approximately 1:2, which indicates that **12** is bivalent for DGL binding. The two DGL binding sites on **12** are the trimannoside moiety on the α(1−6) arm and the α-Man residue on the α(1−3) arm. A similar precipitation profile was obtained with DGL and **14** (data not shown). The precipitation activities of DGL with **12** and **14** are similar to that of ConA (Bhattacharyya et al., 1987a; Bhattacharyya and Brewer, 1989).

Conclusions. The present findings demonstrate very similar but distinct binding properties of DGL and ConA. Both lectins have similar extended binding sites that recognize the same hydroxyl groups in trimannoside **11** which is an epitope found in all N-linked carbohydrates. However, their binding specificities diverge toward larger N-linked carbohydrates including the Man9 oligomannose glycopeptide (**14**) and, more dramatically, toward complex carbohydrates **15** and **33**. DGL is useful in selectively binding to oligomannose carbohydrates since ConA recognizes both oligomannose and complex carbohydrates.

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